

Marginal PMF Explanation



Act as a technical scribe to create exam reference material from the attached slides. Your task is to extract every definition, proof, and worked example, outlining the sequential logic used in each. You must adhere strictly to the provided context—do not introduce new concepts or explanations. Focus on making the transition between logical steps clear and easy to follow for revision purposes.

Here is a **clean, exam-oriented technical scribe version** of your slides, strictly extracted and structured from with **step-by-step logical flow preserved**.

Lecture 18: Marginal PMF, Correlation, Covariance, Joint Gaussian RVs

1. Marginal PMF

Definition

For discrete random variables X and Y :

- Joint PMF:

$$p_{XY}(x, y) = P(X = x, Y = y)$$

- Marginal PMFs:

$$p_X(x) = \sum_y p_{XY}(x, y)$$

$$p_Y(y) = \sum_x p_{XY}(x, y)$$

Key Idea

- The marginal PMF of one variable is obtained by **summing the joint PMF over all possible values of the other variable**. (Page 6)

Marginal PMF from Joint PMF Table

From the table (page 7–8):

- Row sums $\rightarrow p_X(x)$
- Column sums $\rightarrow p_Y(y)$

$$p_X(x_i) = \sum_j p_{ij}, \quad p_Y(y_j) = \sum_i p_{ij}$$

Also:

$$\sum_i p_X(x_i) = 1, \quad \sum_j p_Y(y_j) = 1$$

Worked Example (Step-by-Step)

Joint PMF table (page 9):

	y=0	y=1
x=0	1/8	1/4
x=1	3/8	1/4

Step 1: Compute $p_X(x)$

$$p_X(0) = \frac{1}{8} + \frac{1}{4} = \frac{3}{8}$$

$$p_X(1) = \frac{3}{8} + \frac{1}{4} = \frac{5}{8}$$

Step 2: Compute $p_Y(y)$

$$p_Y(0) = \frac{1}{8} + \frac{3}{8} = \frac{1}{2}$$

$$p_Y(1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

Final Result

$$p_X(x) = \begin{cases} 3/8, & x = 0 \\ 5/8, & x = 1 \end{cases} \quad p_Y(y) = \begin{cases} 1/2, & y = 0 \\ 1/2, & y = 1 \end{cases}$$

2. Correlation

Definition

- Correlation (raw measure):

$$R_{XY} = E[XY]$$

Interpretation

- Measures **average value of the product XY** (page 12)
- Depends on:
 - Mean values
 - Spread
 - Dependence (page 13)

Limitation

- Computed about the origin, not about means
- Not a clean measure of dependence

Transition

To remove mean effects → **Covariance**

3. Covariance

Definition

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Interpretation

- Subtract means first
 - Then measure joint variation (*page 14*)
 - Positive covariance → same direction
 - Negative covariance → opposite direction
 - Zero covariance → no linear relation (*page 15*)
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Algebraic Derivation (Step-by-Step)

Start:

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Expand:

$$= E[XY - X\mu_Y - Y\mu_X + \mu_X\mu_Y]$$

Apply expectation:

$$= E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X\mu_Y$$

Substitute:

$$\mu_X = E[X], \quad \mu_Y = E[Y]$$

Final Result:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Properties (Page 20–23)

1. Symmetry

$$\text{Cov}(X, Y) = \text{Cov}(Y, X)$$

2. Variance as Special Case

$$\text{Cov}(X, X) = \text{Var}(X)$$

3. Linearity

$$\text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y)$$

4. Independence

$$X \perp Y \Rightarrow \text{Cov}(X, Y) = 0$$

 Note:

- Zero covariance $\Rightarrow$ uncorrelated
 - Does **not** imply independence
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Example 1 (Step-by-Step)

Given:

$$Y = -2X + 3$$

Step 1: Use covariance formula

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Step 2: Compute expressions

$$XY = X(-2X + 3) = -2X^2 + 3X$$

$$E[Y] = E[-2X + 3] = -2E[X] + 3$$

Step 3: Substitute

$$\text{Cov}(X, Y) = E[-2X^2 + 3X] - E[X](-2E[X] + 3)$$

Step 4: Simplify

$$= -2E[X^2] + 3E[X] + 2(E[X])^2 - 3E[X]$$

$$= -2E[X^2] + 2(E[X])^2$$

Step 5: Use variance identity

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

Final:

$$\text{Cov}(X, Y) = -2\text{Var}(X)$$

Pearson's Correlation Coefficient**Definition**

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Properties (Page 31–32)

- Measures **strength and direction**
- Normalized covariance
- Unit-free

$$-1 \leq \rho_{XY} \leq 1$$

- > 0 : positive linear trend
 - < 0 : negative linear trend
 - $= 0$: no linear correlation
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Example 2 (Continuous Case)

Given joint PDF:

$$f_{XY}(x, y) = \begin{cases} \frac{xy}{96}, & 0 < x < 4, 1 < y < 5 \\ 0, & \text{otherwise} \end{cases}$$

Step 1: Compute $E[X]$

$$E[X] = \int_0^4 x f_X(x) dx = \frac{1}{8} \int_0^4 x^2 dx = \frac{8}{3}$$

Step 2: Compute $E[Y]$

$$E[Y] = \int_1^5 y f_Y(y) dy = \frac{1}{12} \int_1^5 y^2 dy = \frac{31}{9}$$

Step 3: Compute $E[XY]$

Since X, Y independent:

$$E[XY] = E[X]E[Y] = \frac{8}{3} \cdot \frac{31}{9} = \frac{248}{27}$$

Step 4: Compute $E[2X + 3Y]$

$$E[2X + 3Y] = 2E[X] + 3E[Y] = \frac{16}{3} + \frac{31}{3} = \frac{47}{3}$$

Step 5: Compute $Var(X)$

$$E[X^2] = \frac{1}{8} \int_0^4 x^3 dx = 8$$

$$Var(X) = 8 - \left(\frac{8}{3}\right)^2 = \frac{8}{9}$$

Step 6: Compute $Var(Y)$

$$E[Y^2] = \frac{1}{12} \int_1^5 y^3 dy = 13$$

$$Var(Y) = 13 - \left(\frac{31}{9}\right)^2 = \frac{92}{81}$$

Step 7: Compute Covariance

$$Cov(X, Y) = E[XY] - E[X]E[Y] = 0$$

Step 8: Compute Correlation

$$\rho_{XY} = 0$$

4. Jointly Gaussian Random Variables

Definition

- A pair (X, Y) is jointly Gaussian if:
 - Joint density is **bivariate Gaussian**
 - Marginals are Gaussian (*page 50*)

Joint Gaussian PDF

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(\frac{-1}{2(1-\rho^2)} \left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) \right]\right)$$

Where:

- μ_X, μ_Y : means
- σ_X, σ_Y : standard deviations
- ρ : correlation coefficient (*page 52*)

Interpretation (Page 53–54)

- Models variables influenced by many random processes
- Example:
 - X : CO₂ concentration
 - Y : temperature
- If:

$$\rho > 0$$

→ Higher CO₂ associated with higher temperature

Final Summary Flow (for Revision)

1. Joint PMF → sum → Marginal PMF
2. Raw correlation $E[XY]$ → not centered
3. Center variables → Covariance
4. Normalize covariance → Pearson coefficient
5. Extend to continuous → integrals
6. Special case → Joint Gaussian distribution

If you want, I can convert this into 1-page revision sheet or handwritten-style notes for exam.